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## Integrated Driver Rostering Problem in Public Bus Transit

Lin Xie<sup>a,\*</sup>, Natalia Kliwer<sup>b</sup>, Leena Suhl<sup>a</sup><sup>a</sup>University of Paderborn, Warburger Str.100, 33098 Paderborn, Germany<sup>b</sup>Freie Universität Berlin, Garystr.21, 14195 Berlin, Germany

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### Abstract

The driver rostering problem (DRP), arising in public bus transport companies, generates for each group of drivers a cyclic roster while management considerations, labor laws, and the preferences of drivers have to be satisfied. Optimal rosters are characterized by maximal satisfaction of drivers, minimal difference of overtime among all drivers, and minimal number of unassigned duties. The DRP is mostly solved sequentially due to its high complexity, namely firstly the rota scheduling problem, and secondly the duty sequencing problem. However, this method may generate sub-optimal rosters. In order to avoid a sub-optimal solution, the paper discusses an integrated DRP, which is solved for real-world instances and compared with the results of the sequential approach.

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**Keywords:** integrated driver rostering problem; network optimization model; public transport; rotation-based rostering; simulated annealing

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### 1. Introduction

Traditionally, in public bus transport the driver rostering problem (DRP) is solved after solving a driver scheduling problem. The driver scheduling problem aims at generating anonymous duties while some constraints have to be satisfied, such as the regulation of length and frequency of breaks within a duty. Each duty has its beginning time, ending time, the weekday it belongs to, the shift type (e.g. early shift) etc. Each driver can execute one duty per day. The DRP addresses the task of assigning the duties to drivers while management considerations, labor laws, and the preferences of drivers have to be satisfied. After solving this problem, a roster is generated for each driver. Optimal rosters are characterized by maximal satisfaction of drivers, minimal difference of overtime among all drivers, and minimal number of unassigned duties.

In this paper, we consider *cyclic rosters* which are generated for groups of drivers. A cyclic roster reflects the fairness in the following context: drivers within a group have the same duties including unpopular duties. More

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\* Corresponding author. Tel.: +49-5251-60-5241; fax: +49-5251-60-3542.

E-mail address: [xie@dsor.de](mailto:xie@dsor.de).

details can be found in Section 2. In the remainder of the paper, for simplicity, the word roster refers to cyclic roster. Besides the cyclic roster, there are other ways to generate a roster. A *personalized roster* is generated for a driver and considers wishes with respect to special single free days or vacations. The personalized rostering approach is mostly used in the airline industry, especially in European airlines because of fairness (Kohl and Karisch (2004), Cappanera and Gallo (2004), Maenhout and Vanhoucke (2010)). In recent years, this way of generating roster is also investigated in railways (De Pont (2006)). In *bidline rostering*, which is used for example at North American airlines, a roster is generated subsequently for each individual crew member (see Kohl and Karisch (2004) for an overview).

Traditionally, in public transport the DRP is solved sequentially due to its high complexity: *rota scheduling* first and *duty sequencing* second (Sodhi and Norris (2004), Hartog et al. (2009)). The rota scheduling concentrates on assigning a set of shifts, free days, and reserve shifts to driver groups (rotas are generated) while the duty sequencing problem aims to assign the duties to driver groups according to the optimal rotas. In the sequential DRP, sub-optimal rosters can be generated even though rotas are optimal. In order to avoid the sub-optimality, this paper discusses an integrated DRP, which is solved for real-world instances and compared with the results of the two sequential problems. To the best of our knowledge, this paper presents the first approach to solve the two sequential problems simultaneously in public transit.

The outline of the remainder of this paper is as follows: Section 2 describes the rotation-based crew rostering problem in detail with an example. This example is used in the whole paper. In Section 3 we propose a new network model for the integrated DRP and give a mathematical formulation. This formulation is solved with a standard commercial solver. An improvement approach based on the simulated annealing method is described in Section 4. In order to compare the results of the integrated solution approach, a sequential approach is described shortly in Section 5. Finally, in Section 6 we present computational results showing that our integrated approach outperforms the sequential approach with regard to solution quality.

## 2. Problem description

In rotation-based DRP, desired rota for each driver group and a set of duties are considered as input data for the DRP. As mentioned before, each duty belongs to exactly one shift type depending on its starting time and ending time. *Shift types* can be roughly classified as early, midday and late shifts. However, some German bus companies define shift types more precisely, like very early shift (FFD), early shift (SFD), earlier midday shift (FMD), midday shift (SMD), earlier night shift (FSD) and night shift (SSD) etc. A sequence of consecutive shift types is called *Shift pattern*. Under a *backward shift pattern* the drivers would like to have a late shift at the beginning of a working week, then midday shifts, and an early shift at the end of the working week, in order to get longer free days before and after the working week, and vice versa for a *forward shift pattern*. A *steady shift pattern* contains the same or similar shifts within a working week.

A *desired rota* is a sequence of desired shift patterns, free days, reserve duties as well as some pre-assigned activities for a group of drivers. Table 1 shows an example of a desired rota for five weeks. The drivers are gathered in *groups* according to the same qualification, such as a particular vehicle license, same route knowledge, and same desires of the shift pattern (backward, forward, steady). The number of weeks is equal to the number of drivers in a roster, so each driver gets a different row. The first driver begins with the first row in first week while the second driver begins with the second row, etc. After finishing the duties in the actual row, the drivers begin with the duties in the subsequent row. The duties in the first row follow the duties in the last row. When a driver has accomplished all duties of the rota, he or she begins the duties of the first week again. This explains the word 'rotation'.

Table 1. An example of a desired rota

| Week | Mon | Tue | Wed | Thu | Fri. | Sat | Sun |
|------|-----|-----|-----|-----|------|-----|-----|
| 1    | R   | R   | R   | R   | R    | F   | F   |
| 2    | SSD | FSD | SMD | FMD | SFD  | F   | F   |
| 3    | SSD | FSD | SMD | FMD | SFD  | F   | F   |
| 4    | SSD | FSD | SMD | FMD | SFD  | F   | F   |
| 5    | SSD | FSD | SMD | FMD | SFD  | F   | F   |

Table 2. An example of possible admissible shifts with the corresponding costs

|   | Shift1 | Shift2 | Cost |
|---|--------|--------|------|
| 1 | SFD    | FFD    | 50   |
| 2 | SFD    | FMD    | 100  |
| 3 | FMD    | SMD    | 50   |
| 4 | SMD    | FMD    | 50   |
| 5 | FSD    | SSD    | 50   |
| 6 | SSD    | FSD    | 50   |

Table 3. An example of admissible successor shifts with the corresponding costs

|   | Shift1 | Shift2 | Cost |
|---|--------|--------|------|
| 1 | SSD    | FSD    | 0    |
| 2 | SSD    | SSD    | 100  |
| 3 | FSD    | SMD    | 0    |
| 4 | FSD    | FMD    | 100  |
| 5 | SMD    | FMD    | 0    |
| 6 | SMD    | SMD    | 100  |
| 7 | FMD    | SFD    | 0    |
| 8 | FMD    | FFD    | 50   |
| 9 | FMD    | FMD    | 100  |

One of the objectives for the solving rotation-based DRP is to maximize the number of desired shifts. Therefore, undesired shifts are gradually punished. For example, another shift type as a desired early shift (SFD) is punished, but a very early shift (FFD) is less punished than earlier midday shift (FMD) (see Table 2 row 1 and 2). Therefore, all admissible alternatives of a shift (Table 2) as well as admissible successors (Table 3) with the corresponding cost are defined. Note that a desired successor has the cost 0. The costs shown in Table 2 and 3 are artificial costs used to satisfy the desires as far as possible.

After solving the rotation-based DRP, a roster is generated for each group of drivers. A *roster* is a sequence of duties, free days as well as other activities, such as reserve duties and training, from Monday to Sunday for several weeks. To ensure the generation of a feasible roster, some hard regulations must be held during generation, such as rest periods between two duties, maximal consecutive working days, maximum number of single day-off etc.

### 3. Integrated Approach

In this section we discuss our solution approach for the integrated DRP. First of all, the free days and reserve duties are pre-assigned based on a given desired rota (see Section 3.1). In Section 3.2 a desired shift-block network is described for the rest of the desired rota. For each desired shift-block networks a duty-block is defined. We describe the duty-block network and the mathematical formulation in Section 3.3 and 3.4.

#### 3.1. Pre-assignment of free days and reserve duties

The desired distributions of reserve duties and free days are checked for feasibility first. The goal is to retain the distribution to the largest possible extent, unless the following hard constraints cannot be satisfied.

- The minimum/maximum number of consecutive working days
- The maximum number of consecutive free days
- The minimum and maximum number of single/two consecutive free days
- The distance between two consecutive free days should be less than 14 days

The same mathematical formulation for the restrictions can be found in Emden-Weinert et al (2000) and Speer (2000).

The distributions of reserve duties are currently evenly planned for all drivers without considering more detailed information such as historical or weekday-dependent sickness absence rates. A stochastic model for better planning of reserve duties is presented in Xie et al. (2012).

#### 3.2. Desired shift-block network

After pre-assignment of free days and reserve duties (F and R in Table 2), possible duties are assigned to the rest of desired rota. We define the term *desired shift-block* as a sequence of consecutive desired shifts (e.g. in Table 1: SSD→FSD→SMD→FMD→SFD from Monday to Friday) in a desired rota. Note that the length of a block should be less than or equal to the maximal number of consecutive working days. The similar desired shift-blocks are gathered to determine the frequency, and we generate a desired shift-block network for all of them. (e.g., in Table 1: the frequency of SSD→FSD→SMD→FMD→SFD from Monday to Friday is equal to four). Two blocks are *similar*, if they have the same shift pattern, and each of the shift types in the shift pattern is carried out on the same day. Based on the given possible alternative and successor shifts described in the example in Section 2, desired shift-block network is generated as follows (see Figure 1). Each *node* in the network represents either the desired shift type or an alternative shift type on a weekday. An *arc* between two nodes indicates that the combination of two nodes is possible (based on Table 3). A shift type of the first day without a successor or a shift type of the last day without a predecessor will be eliminated (e.g. FSD on Monday as an alternative shift for SSD). A shift type of the rest of the days will also be eliminated, if it does not have any successor or predecessor (e.g. SSD on Tuesday as an alternative shift for FSD).

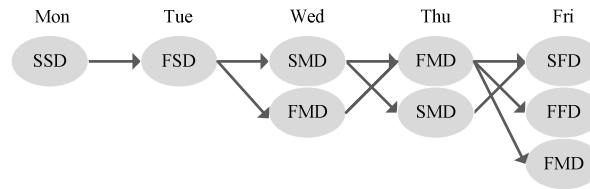


Fig. 1. The desired shift-block network of previous example in Section 2

### 3.3. Duty-block Network

A *duty-block* is defined as sequences of all possible consecutive duties in a desired shift-block network, which is described in the previous section. We denote the set of possible shift types of a day  $t$  by  $S_t$ , which can be generated from the desired shift-block network. The possible duties of a shift type  $s$  on a day  $t$  would be collected to a set  $D_{s,t}$ . Let the frequency of the desired shift-block be  $f$ .

For each duty-block a network is generated in the following way. A *Node* in the network is a starting node  $s$  or an ending node  $t$  of the network or a beginning or ending node of a duty/dummy duty. The starting node  $s$  is connected to all duties  $d \in D_{s,1}$  with  $s \in S_1$  and the dummy duty of the first day. The ending node  $t$  is connected to all duties  $d \in D_{s,5}$  and  $s \in S_5$  and the dummy duty of the last day. A *duty arc* represents a duty  $d \in D_{s,t}$  with  $s \in S_t$  and has at least one incoming and one outgoing arc, i.e. the upper bound is 1. A *dummy arc* is added for each day once to guarantee a feasible solution. The upper bound of this dummy arc is set to  $f$ . A *circulation arc* from  $t$  to  $s$  is added to guarantee that exactly  $f$  paths are generated and its lower and upper bound are set to  $f$ .

The connections between those arcs are needed to complete the network. Thus, a *connecting duty arc* is added where an arc is connected with duty arcs, so its upper bound is equal to the upper bound of a duty arc, i.e. one. Two duty arcs  $d1$  and  $d2$  can only be connected, if they are compatible. *Compatible* means that  $d2$  is a possible successor of  $d1$  (see Section 2) and the rest period between two duties is larger as the defined overtime rest period (e.g. 11 hours). A *connecting dummy arc* is only added between circulation arc and dummy arc or between dummy arcs, and its upper bound is equal to  $f$ .

Figure 2 depicts a duty-block network for the example which is shown in previous sections. We assume that there are four possible duties on each day.

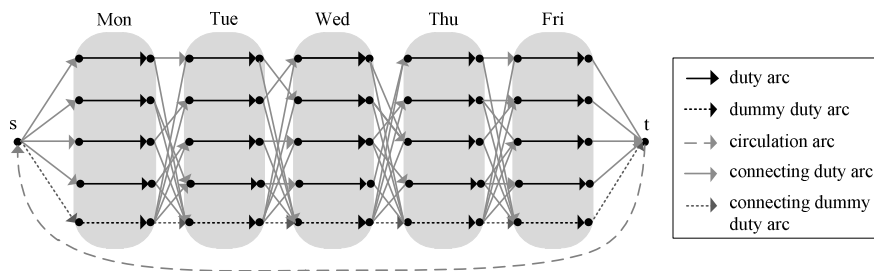


Fig. 2. The duty-block network of previous example in Section 2

### 3.4. Mathematical Formulation

In this section we present our mathematical formulation, which relies on the network structure described in Section 3.3. Let  $D = \{1, 2, \dots, n\}$  be the set of  $n$  duties and  $B = \{1, 2, \dots, m\}$  be the set of  $m$  duty-blocks. For each duty-block  $b \in B$ ,  $G^b = (N^b, E^b)$  defines a cyclic network as described earlier, with  $N^b$  as the set of nodes and  $E^b$  as the set of edges. We denote  $DE^b \subset E^b$  as the set of duty arcs. The set of connecting duty arcs is denoted as  $CDE^b \subset E^b$ . Let  $DE^d$  be the set of arcs representing duty  $d$  in all duty-blocks. We introduce the dummy duties to ensure a feasible solution. Let  $DUE^b \subset E^b$  be the set of dummy arcs, and the set of connecting dummy arcs is denoted as  $CDUE^b \subset E^b$ . Binary decision variables  $x_e$  with  $e \in DE^b \cup CDE^b$  indicate whether the edge  $e$  is selected or not. Besides that, we use linear decision variables  $x_e$  with  $e \in DUE^b \cup CDUE^b$  indicating how many times the edge  $e$  can be selected. All linear decision variables have a maximum capacity  $f^b$  as described in the previous section. With each arc  $e \in E^b$  we associate a cost  $c_e$  that is the total cost of adding the arc  $e$ . The cost includes the alternative or successor cost of adding that arc. The integrated crew rostering problem with multiple duty-blocks can be stated as follows:

$$\min \sum_{b \in B} \sum_{e \in E^b} c_e * x_e \quad (3.1)$$

$$\sum_{i|(i,j) \in E^b} x_{ij} - \sum_{i|(j,i) \in E^b} x_{ji} = 0 \quad \forall i \in N^b, b \in B \quad (3.2)$$

$$\sum_{e \in DE^d} x_e \leq 1 \quad \forall d \in D \quad (3.3)$$

$$x_e \in \{0,1\} \quad \forall e \in DE^b \cup CDE^b, b \in B \quad (3.4)$$

$$0 \leq x_e \leq f^b \quad \forall e \in DUE^b \cup CDUE^b, b \in B \quad (3.5)$$

$$x_{e=(t,s)^b} = f^b \quad \forall b \in B \quad (3.6)$$

The objective (3.1) minimizes the sum of costs, i.e. it maximizes the satisfaction of drivers. Constraints (3.2)-(3.3) correspond to a multicommodity flow formulation for the duty-network crew rostering problem where flow conservation is ensured for each duty-block (3.2) and each duty can only be assigned for all duty-blocks at most once (3.3). Due to restriction (3.3) it may have some unassigned duties in the solution, the number of which should be minimized. Although this objective does not explicitly exist in (3.1), the costs of selected dummy arcs and connecting dummy duty arcs are set to a large number. It induces to minimize the number of dummy arcs. In other words, it minimizes the number of unassigned duties. Constraint (3.4) defines that each duty arc or connecting duty arc corresponds to a binary variable, while constraint (3.5) guarantees that the capacity of the dummy duty or connecting dummy duty arc is less than  $f^b$ . Constraint (3.6) ensures that there are exactly  $f^b$  paths in the solution.

#### 4. Simulated Annealing for improvement

The mathematical model in Section 3.4 does not consider the minimization of the variation of overtime among all drivers as an objective. Emden-Weinert et al. (2000) used simulated annealing to solve the sub-problem of DRP, namely the duty sequencing problem. In this work, we use a similar simulated annealing approach to improve the results of the integrated rostering problem described in Section 3. Emden-Weinert et al. (2000) relaxed the constraints that forbid any inadmissible duties, so that a possible initial solution could be found. The post-processing was needed to find all inadmissible duties. The initial solution of our simulated annealing approach is the optimal solution of the network model described in the previous section. We do not allow any inadmissible duties, but dummy duties are possible in the initial solution. The maximum of all weekly overtimes

in all driver groups can be minimized while minimizing the difference of average overtime between all groups. The two objectives ensure that all drivers have similar overtimes.

The basic procedure is shown in Figure 3. In the initialization step, the solution of the integrated model described in previous section  $s_0$  is set to be the initial solution, and the number of iterations, the initial temperature  $t_0$  as well as cooling function  $\alpha$  is selected.  $s \in N(s_0)$  denotes a feasible neighbor of  $s_0$ , i.e. one duty  $d1$  is switched with another one  $d2$ , which has the same shift type and holds the restrictions that  $d1$  is compatible with the predecessor and successor of  $d2$ , and vice versa. If a better solution can be generated by switching duties ( $\delta < 0$ ), then store  $s$  to  $s_0$  and reduce the temperature. Otherwise, a randomly generated number  $x$ , which is between 0 and 1, will be tested, whether it is less than  $\exp(-\delta/t)$ . If this is the case, then we still accept this worse solution. Otherwise, we should find another duty within the same temperature till the defined number of iterations is reached.

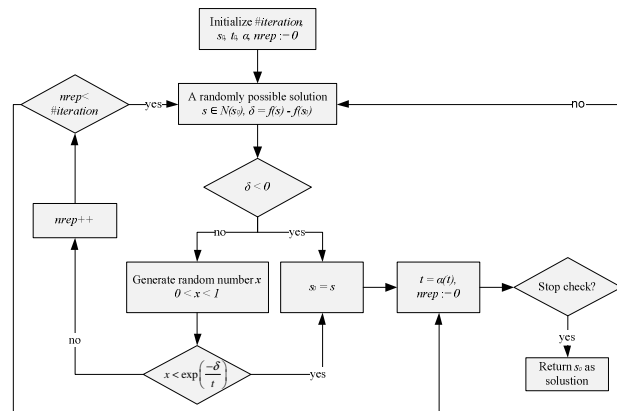


Fig. 3. General process of simulated annealing

## 5. The sequential approach

The sequential approach we use in this paper is similar to the integrated approach presented in Section 3 in the following way. The sequential DRP is divided into two steps, as mentioned in Section 1, namely the rota scheduling problem and the duty sequencing problem. In rota scheduling, the day-offs and reserve duties are pre-assigned (as described in Section 3.1), and the desired shift networks according to Section 3.2 are generated. A set of possible shift blocks can be enumerated based on a desired shift-block network. Recall that a shift block means a sequence of consecutive shifts. Let  $D = \{1, 2, \dots, m\}$  be the set of  $m$  shift types and  $N = \{1, 2, \dots, n\}$  be the set of  $n$  desired shift-block networks.  $T = \{1, 2, \dots, 7\}$  is the set of week days. We denote  $S^n$  be the set of shift-blocks in network  $n \in N$ . The number of shift types  $m \in D$  on day  $t \in T$  is defined as  $a_{m,t}$ . Let  $c(s, m, t)$  be a function checking whether shift-block  $s \in S^n$  has shift type  $m \in D$  on day  $t \in T$ .  $b_n$  is defined as the number of shift-blocks that should be assigned in network  $n$ . Linear decision variable  $x_s$  with  $s \in S^n$  and  $n \in N$  indicates how many times the shift-block  $s$  in network  $n$  is assigned. Its upper-bound is defined as  $b_n$ . With each shift-block  $s \in S^n$  we associate cost  $c_s$  that is the total cost of adding shift-block  $s$  in the solution which was calculated during the generation of the shift-block. The rota scheduling problem can be stated as follows:

$$\min \sum_{n \in N} \sum_{s \in S^n} x_s * c_s \quad (5.1)$$

$$\sum_{s \in S^n} x_s = b_n \quad \forall n \in N \quad (5.2)$$

$$\sum_{n \in N} \sum_{s \in S^n} c(s, m, t) * x_s \leq a_{m,t} \quad \forall m \in D, t \in T \quad (5.3)$$

$$x_s \in \{0, \dots, b_n\} \quad \forall n \in N, s \in S^n \quad (5.4)$$

The objective (5.1) minimizes the sum of costs, i.e. it maximizes the satisfaction of drivers. Constraint (5.2) guarantees that there are  $b_n$  shift-blocks selected in the solution while constraint (5.3) makes sure the sum of all shift types in the selected shift-blocks does not exceed the daily requests. Due to restriction (5.3), each day a dummy duty is added to the desired shift-block network in order to obtain a solution (see Figure 4). The dummy duties are selected with high costs. Therefore, they will be minimized, due to the objective function (5.1). Constraint (5.4) defines the variables.

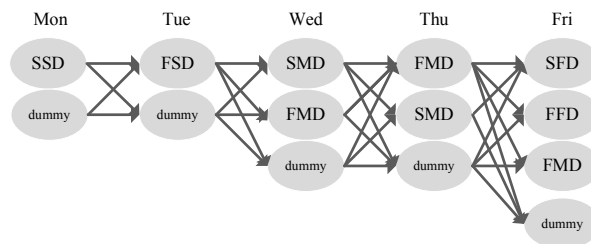


Fig. 4. Extended desired shift-block network based on Figure 1

The selected shift-blocks are used as input for duty generation network described in Section 3.3. The same mathematical formulations shown in Section 3.4 are also suitable for solving the duty sequencing problem.

## 6. Computational Results

In this section we report computational results of our modeling and solution approach. All experiments were conducted on a Dell Latitude E5510 personal computer running Windows 7 64 bits with an Intel Core i5 processor @2.53GHz and 4 GB of main memory. Our integrated and sequential models shown in Section 3 and 5 are solved by the MIP solver ILOG CPLEX 12.0.

We tested our approaches on two real-world medium-sized instances from public transport companies, namely 256\_1013\_9 and 256\_899\_9. The instance name contains information about the problem data: The number of drivers, the number of duties, and the number of different driver groups.

The parameter we used for running the both instances are defined as follows:

#iteration is set to 200,

the temperature at the beginning is set to 500,

$\alpha$  is set to 0.95.

As shown in Table 4, it is worth to use simulated annealing to minimize the *maximal weekly overtime of all drivers* (in short: MWOoD) and the *difference of average overtime between groups* (in short: DoAOG). For both instances, more than 80% of MWOoD is reduced. The values of DoAOG are decreased above 70%. The running time of simulated annealing for each instance is less than 2 minutes.

In Table 5 we report results of our integrated approach, which are compared with the sequential approach based on the following criteria.

The complete running time

The number of unassigned duties.

The number of duties which are assigned exactly as desired.

The number of duties which are assigned but not as desired.

The MWOoD and DoAOG that are obtained by simulated annealing for sequential and integrated approaches.

Note that the running time of our integrated approach is not much longer than the running time of the sequential one. Our integrated models can be solved within about 10 minutes for each instance. The instance 256\_899\_9 can be solved faster compared to the instance 256\_1013\_9, because there are less alternative shifts to the desired shifts in this instance. That is why the number of unassigned duties is higher in this instance, although the number of duties, which are required to be assigned, is smaller. The same reason implies a smaller improvement of the number of unassigned duties than for the instance 256\_1013\_9. There are more duties assigned as desired in the integrated model in both instances. Therefore, the numbers of duties, which are assigned as alternative shifts, are smaller compared to the sequential model. Furthermore, due to the objective function (3.1), 71% of them in instance 256\_1013\_9 as well as 83.5% of them in instance 256\_899\_9 are assigned with pretty low costs (as shown in Figure 5). In other words, those are similar duties compared to the desires of the drivers, e.g., instead of a very early duty, which a driver prefers on Monday, he takes an early duty. The values of MWOoD and DoAOG of the integrated models are smaller or almost equal to the sequential model, especially the values of MWDOD, i.e., the integrated solution provides the distribution of overtimes for all drivers more evenly.

Table 4. The comparison before and after running simulated annealing for integrated model

| criteria        | 256_1013_9 |       |             | 256_899_9 |       |             |
|-----------------|------------|-------|-------------|-----------|-------|-------------|
|                 | before     | after | improvement | before    | after | improvement |
| MWOoD (in min.) | 1022       | 185   | +82%        | 866       | 146   | +83,1%      |
| DoAOG (in min.) | 179.7      | 51.1  | +71.6%      | 402.7     | 52.7  | +86.9%      |

Table 5. The comparison of the results of sequential and integrated approaches

| criteria                          | 256_1013_9 |            |             | 256_899_9  |            |             |
|-----------------------------------|------------|------------|-------------|------------|------------|-------------|
|                                   | sequential | integrated | improvement | sequential | integrated | improvement |
| running time (in sec.)            | 495        | 628        | -26.8%      | 130        | 162        | -24.6%      |
| # unassigned duties               | 59         | 23         | +61.0%      | 124        | 103        | +16.9%      |
| # duties assigned as desired      | 665        | 726        | +11.2%      | 538        | 626        | +16.4%      |
| # duties assigned as alternatives | 289        | 264        | +8.7%       | 237        | 170        | +28.3%      |
| MWOoD (in min.)                   | 224        | 185        | +17.4%      | 193        | 146        | +24.3%      |
| DoAOG (in min.)                   | 52.8       | 51.1       | +3.2%       | 52.5       | 52.7       | -0.38%      |

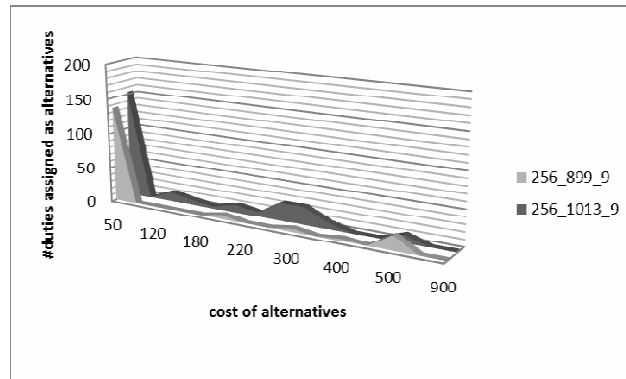


Fig. 5. The distribution of costs of duties in the integrated model, which assigned as alternative shift types

## 7. Summary

In this paper, we have proposed a new modeling approach for the integrated crew rostering problem in public transport. The formulation relies on a new duty-block network. Our results indicated that the medium-sized instances with 256 drivers and 1013 duties / 899 duties can be solved very efficiently. Moreover, our method outperformed a sequential approach in terms of solution quality. About further research we will tackle larger real-world instances.

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